

National University of Computer & Emerging Sciences- FAST

Karachi Main Campus

Department of Computer Science

First Mid Terminal Examination-Fall 2009

MT101- Calculus I

Dated: 16 September 2009

Max Marks: 30

Max Time: 60Min

Attempt all questions:

Show all necessary working and steps:

Q.1 Find the domain and range of the functions

i) $2 + \sqrt{x-1}$

ii) $\frac{1}{1 - \sin x}$

Q.2 Sketch the graph of equation $y = -2(x+1)^2 - 3$

Q.3 Find a formula for $fohog$ where $f(x) = 2 + \sqrt{x-1}$, $g(x) = \frac{x}{1+x^2}$ and $h(x) = 2 - x^2$

Q.4 Find $\frac{dy}{dx}$ where

i) $y = [x \sin 2x + \tan^4(x^7)]^5$ (ii) $x \cos y = y$

Q.5 Evaluate the integral $\int \cos 4\theta \sqrt{2 - \sin 4\theta} d\theta$

Q.6 Use Leibnitz theorem to find 3rd derivative of the given functions.

$(x^4 + 1)\sin x$